

Can Qwen3-VL alone replace the extraction pipeline?

Same 16-page Coulton PDF, current Qwen3-VL 4B, image-only versus the complete OCR / OpenCV / Qwen source-block run.

Answer: not wholesale. Qwen alone reads several charts and legal columns better than today's code, but loses map owners, shifts loan-table cells, and can fail to close JSON. Keep evidence-based verification; let vision propose repairs where it wins.

56 / 65 VISION-ONLY ANCHORS	50 / 65 COMPLETE-PIPELINE ANCHORS	15 / 16 VISION JSON COMPLETION	10.2 / 4.2 MINUTES: VISION / FULL
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Selected content-and-owner checks

PAGE / TASK	VISION	FULL	KEY SIGNAL
3 - KPI amount / count / caption	4/4	0/4	Vision keeps both clusters
6 - Financial table cells	16/18	18/18	Full protects blank-cell columns
7 - One pie: category / percent	10/10	10/10	Both read all 10
8 - US map: state / value	5/10	10/10	Full owns values by geometry
9 - Main + inset pie pairs	4/5	0/5	Vision reads 4; neither owns inset
10 - Offer-lane numeric claims	6/6	6/6	Both read all 6
12 - Scatter / text safety	5/6	4/6	Vision avoids fake 100% point
16 - Legal clauses / column order	6/6	2/6	Vision restores left-then-right
Selected total	56/65	50/65	One-PDF development score

The 65 anchors are heterogeneous and intentionally selected from eight hard pages. This is a descriptive development score, not a blinded multi-document accuracy estimate.

What the new model gets right - and wrong

The older brief tested Qwen3.5:4B/9B, not this Qwen3-VL 4B instruct profile. The two score endpoints must not be pooled.

Strengths worth using

Pages 3, 7 and 9: Vision-only gets all four KPI cluster anchors, all ten simple-pie labels, and four of five amount/percent pairs in two linked pies. The full run is wrong on page 3 labels and splits page 9 into 16 observations totaling 167.7%. Qwen still omits the 0.7% Return of Capital and does not make a real main/inset chart hierarchy.

Pages 11, 13 and 14: On printed bar labels, vision-only and full output agree exactly on 10/10, 10/10 and 23/23 pairs respectively. This is output agreement, not independent truth. **Page 16:** Vision-only reads the legal columns in the correct left-then-right order; current full text interleaves them while many blocks still claim `passed`.

Where source ownership breaks

CASE	OBSERVED FAILURE
PAGE 6 / TABLE	72/84 data cells agree on value <i>and column</i> ; 12 differ. Qwen shifts \$28,045,777 from Non-controlling Interest into Leverage and moves \$27,598,623 left. Native-PDF cell geometry catches this.
PAGE 8 / MAP	Qwen calls the map a chart and emits 50 map rows against 27 full bindings. Only 7/50 agree; 35 repeat New York, generally with 1.3%. Geography registration and leader lines materially constrain owners.
PAGE 5 / OUTPUT	First-pass JSON was 15/16. A 155.5-second repeated photo description ended mid-string; the same temperature-zero retry failed. A compact-prompt probe also failed on pages 2 and 4.

Speed and contract

Image-only first attempts: **609.4 seconds** for 16 pages attempted (including failed page 5). Saved full pipeline: **254.5 seconds** for 16 pages complete, about 2.39x faster in these separate sessions. The extra 154.7-second image-only retry is excluded. The full output validates all 149 blocks structurally, but its semantic `passed` flag does not catch page 16 column corruption.

Image-only JSON used a deliberately lighter semantic-page contract. Exact content/owner anchors are compared head-to-head; evidence IDs, source coordinates, hash-backed crops, per-page inspection and unified hierarchy remain additional deliverables of the complete architecture, not fields that Qwen was asked to produce.

A practical step-one decision

Rahul's model-first challenge exposed both genuine vision gains and unguarded failure modes.

Recommended division of labor

1. Literal evidence

Keep native PDF text / RapidOCR for printed words, dates, numbers and negation; compare Qwen claims to visible tokens.

2. Structural ownership

Keep PDF/native table cells, OpenCV/vector plot geometry and registered map boundaries where blank cells, wedges, axes and states decide the owner.

3. Vision candidate

Use current Qwen3-VL routinely for chart, KPI, comparison and column hypotheses. Accept its better page 3/9/16 readings when independent evidence supports them.

4. Fail closed

Reject repeated owners, unsupported values, axis ticks as data, malformed JSON and missing chart hierarchy. Mark unresolved items for human review.

Fix the complete pipeline, too

General rules still needed: page 3 KPI amount/count/caption clustering; page 9 multiple-pie segmentation and per-pie denominators; page 12 scatter axis-vs-data and point-completeness checks; page 16 column-lane reading order with a cross-column validator. Page 6's non-native PaddleOCR branch remains unimplemented and was not exercised in this PDF.

What this benchmark cannot decide

This is one development PDF and a visibly checked 65-anchor reference made with knowledge of previous runs, not a blind human gold set. Model-only receives full page images; full extraction uses crops, OCR and geometry. Runtime is not hardware-randomized. The chosen anchor mix cannot be turned into a production F1 score or a statistically general model ranking. A blind multi-document test with scanned tables, complex maps and multi-column legal pages is the required architecture gate.

Reproduce and inspect

Repository folders: **reports/vision-ablation-qwen3vl/** (reference, scoring code, detailed Markdown) and **benchmarks/qwen3vl-vision-only-coulton-20260915-r3/** (per-page JSON, raw text, receipts, failures and score). Comparator: **examples/coulton-creek-v070-full-qwen-audited-r1/**. Source PDF SHA-256: cf569a8b25fcd36301b34e0e976e88227faa9f9cdc7804f34eddad09e8d477ad.